

PRATEET TIWARI

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EXPERIENCE

Frontend Engineer Intern | [TechSnap](#)
Bengaluru, India (Remote)

Mar 2025 – May 2026

- Developed high-performance dashboards using **Next.js**, improving Core Web Vitals and SEO performance.
- Optimized client-side data management with **TanStack Query** and **Redux Toolkit**, reducing redundant API requests and improving application responsiveness.

PROJECTS

[StreamIt - Video Streaming Platform](#) | *Go, Gin, Redis, AWS S3, FFmpeg, TypeScript, Node.js, Next.js*

- Built a distributed video processing pipeline using **Go**, **Asynq**, and **FFmpeg** workers for asynchronous transcoding and media processing.
- Implemented adaptive bitrate **HLS** streaming with parallel transcoding using **goroutines** and semaphore-based concurrency control.
- Architected a distributed view-counting pipeline using **Redis Streams**, consumer groups, heartbeat validation, deduplication, and **HyperLogLog**, with batched counter synchronization to **MongoDB**.

[Sentinel - Distributed API Gateway](#) | *Go, gRPC, Protocol Buffers, SQLite*

- Built a **high-performance API gateway** in Go with Radix routing, rate limiting, load balancing, health checks, circuit breaking, retries, and reverse proxying.
- Designed a decoupled **Control Plane/Data Plane architecture** using **gRPC** streaming, SQLite persistence, atomic runtime configuration, and automatic control-plane reconnection without database access on the HTTP hot path.
- Implemented **fault-tolerant traffic management** with full-jitter retries, request-body replay safety, and concurrency-safe components; benchmarked at **52K req/s** with 100 concurrent connections.

[Celestial - Event-Driven HTTP Server](#) | *C, kqueue, HTTP/1.1, Makefile*

- Built an event-driven HTTP server from scratch in **C** using non-blocking sockets and **kqueue** for I/O mux.
- Implemented the **HTTP/1.1 request lifecycle**, including connection management, HTTP parsing, request handling, and response generation.
- Designed a state-driven event loop to handle multiple concurrent TCP connections without blocking, with low-level socket and memory management.

OPEN SOURCE

[Cal.com](#) | Fixed App Store search inconsistencies through input normalization ([PR #29236](#))

[Directus](#) | Fixed incorrect pie chart tooltip values, improving analytics accuracy ([PR #27356](#))

TECHNICAL SKILLS

- **Languages:** Go, TypeScript, JavaScript, Python, C
- **Frameworks & Libraries:** Node.js, Gin, Next.js, React.js, NestJS, TanStack Query, Redux Toolkit, Socket.IO
- **Databases & Storage:** PostgreSQL, MongoDB, Redis, SQLite, AWS S3
- **Infrastructure & DevOps:** Docker, Nginx, Git, GitHub Actions, Vercel, Render

EDUCATION

ABES Engineering College, Ghaziabad, India
Bachelor of Technology in Computer Science and Engineering (CSE)

Expected Aug 2027